are due to differences in the structure and composition of the cell wall. The difference between Gram Positive and Gram negative bacteria is given in Table 1.6.

What are Magnetosomes ?

Intracellular chains of 40-50 magnetite (Fe_3O_4) particles are found in bacterium *Aquaspirillum magnetotacticum*. and it help the bacterium to locate nutrient rich sediments.

1.4.5 Life processes in Bacteria

Respiration

Two types of respiration are found in Bacteria. They are 1. Aerobic respiration 2. Anaerobic respiration.

1. Aerobic respiration

These bacteria require oxygen as terminal acceptor and will not grow under anaerobic conditions. (i.e. in the absence of O_2) **Example:** *Streptococcus.*

Obligate aerobes

Some *Micrococcus* **species** are obligate aerobes (i.e. they must have oxygen to survive).

2. Anaerobic respiration

These bacteria do not use oxygen for growth and metabolism but obtain their energy from fermentation reactions.**Example:** *Clostridium.*

Facultative anaerobes

There are bacteria that can grow either using oxygen as a terminal electron acceptor or anaerobically using fermentation reaction to obtain energy. When a facultative anaerobe such as *E. coli* is present at a site of infection like an abdominal abscess, it can rapidly consume all available O_2 and change to anaerobic metabolism producing an anaerobic environment and thus allow the anaerobic bacteria that are present to grow and cause disease. **Example:** *Escherichia coli* and *Salmonella.*

Capnophilic Bacteria

Bacteria which require CO_2 for their growth are called as capnophilic bacteria. **Example:** *Campylobacter.*

Nutrition

On the basis of their mode of nutrition bacteria are classified into two types namely autotrophs and heterotrophs.

I Autotrophic Bacteria

Bacteria which can synthesise their own food are called autotrophic bacteria. They may be further subdivided as

A. Photoautotrophic bacteria

Bacteria use sunlight as their source of energy to synthesize food. They may be

1. Photolithotrophs

In photolithotrophs the hydrogen donor is an inorganic substance.

a. Green sulphur bacteria: In this type of bacteria the hydrogen donor is H_2S and possess pigment called **Bacterioviridin.** Example: *Chlorobium.*

b. Purple sulphur bacteria: For bacteria belong to this group the hydrogen donor is thiosulphate, **Bacteriochlorophyll** is present**.** Chlorophyll containing chlorosomes are present Example: *Chromatium*.

2. Photoorganotrophs

They utilize organic acid or alcohol as hydrogen donor. Example: Purple non sulphur bacteria – *Rhodospirillum.*

B. Chemoautotrophic bacteria

They do not have photosynthetic pigment hence they cannot use sunlight energy. This type of bacteria obtain energy from organic or inorganic substance.

1. Chemolithotrophs

This type of bacteria oxidize inorganic compound to release energy.

Examples:

1. Sulphur bacteria - *Thiobacillus thiooxidans*
2. Iron bacteria - *Ferrobacillus ferrooxidans*
3. Hydrogen bacteria - *Hydrogenomonas*
4. Nitrifying bacteria - *Nitrosomonas* and *Nitrobacter*

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